

What ShinkaEvolve finds that hand-coded symmetry does not: a generation-by-generation look at three model-ensemble tiers

Banana Labs

Abstract

We ran ShinkaEvolve, an LLM-driven evolutionary search, on an 8-qubit ansatz design task whose hidden structure is a permutation symmetry, and compared it against a hand-built ansatz that encodes that symmetry directly. Evolutionary search with a frontier-tier model ensemble does not just match the hand-coded baseline; it beats it by a wide margin, and it does so by discovering the symmetry itself, unprompted, from the shape of the feature map alone. Repeating the same search with weaker model ensembles shows this is not a generic property of evolutionary search: the ability to recognize the symmetry, name it, and act on it in a single step is concentrated almost entirely in the strongest model, and it is what separates a run that reaches state of the art from one that plateaus well below a naive hand-coded design.

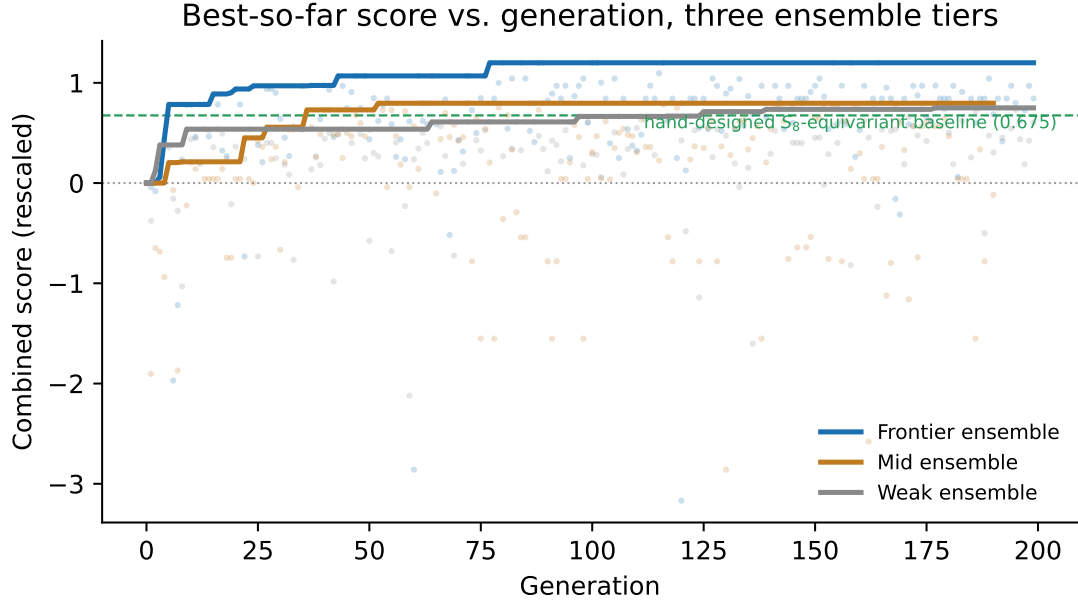
Task

The task, transfer-sn, asks an LLM proposer to evolve an 8-qubit variational block for a 28-feature binary classifier, without telling it what the features are. In fact, the features are the 28 upper-triangle adjacency bits of an 8-vertex graph, the label is graph connectedness, and both the qubit indices and the feature order are scrambled by a fixed random permutation. Relabeling the 8 graph vertices permutes qubits and features jointly and never changes the label, so the task is invariant under the full symmetric group S_8 . Nothing in what the proposer sees names this symmetry; the only trace of it is that the fixed feature map couples every one of the $\binom{8}{2} = 28$ qubit pairs, once each.

Setup

Three ShinkaEvolve runs, 190–200 generations each, differ only in the three-model LLM ensemble proposing mutations: a weak tier (gpt-5-mini, grok-code-fast-1, Phi-4, reasoning disabled), a mid tier (gpt-5.4, Mistral-Large-3, DeepSeek-V4-Pro, reasoning effort xhigh), and a frontier tier (gpt-5.6-sol, DeepSeek-V4-Pro, Mistral-Large-3, reasoning effort xhigh). All three share the same seed, fixed architecture, and scoring rule, so scores are directly comparable across runs. As a fourth reference point we hand-built baseline_program.py: one rotation angle shared across all 8 wires (one gate orbit) and one shared angle across all 28 wire pairs on a complete-graph entangler (the other gate orbit), which is exactly S_8 -equivariant by construction. It scores 0.675 on the run’s own scale (0 = seed, 1 = the previously best evolved solution) with no search at all.

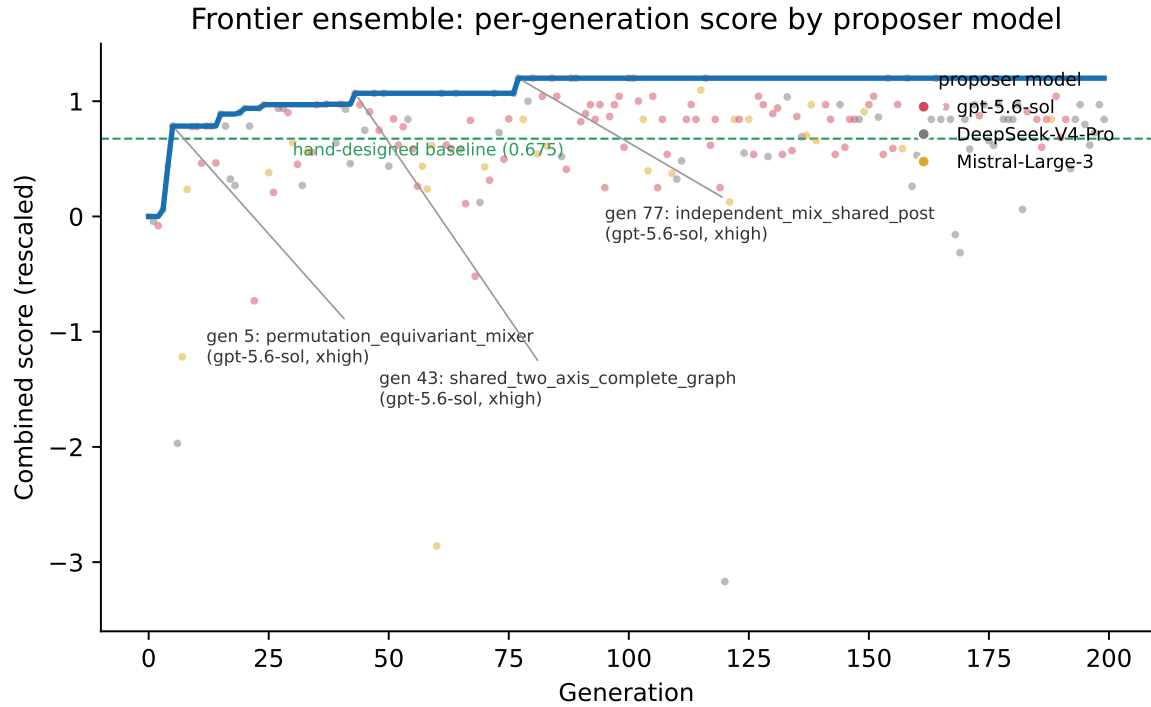
Results



All three runs clear the hand-coded baseline, but by very different margins and at very different speeds. The weak and mid runs plateau just above it (0.750 and 0.796); the frontier run blows past it almost immediately and keeps improving to 1.200, a solution that a human handed only “it’s S_8 -symmetric” would not obviously reach on the first try.

Run	Best score	Best params	Test acc. (best)	Gen. of best
Hand baseline (no search)	0.675	20	0.920	—
Weak ensemble	0.750	20	0.937	177
Mid ensemble	0.796	20	0.930	52
Frontier ensemble	1.200	20	0.980	77

The frontier run in detail: generation 5

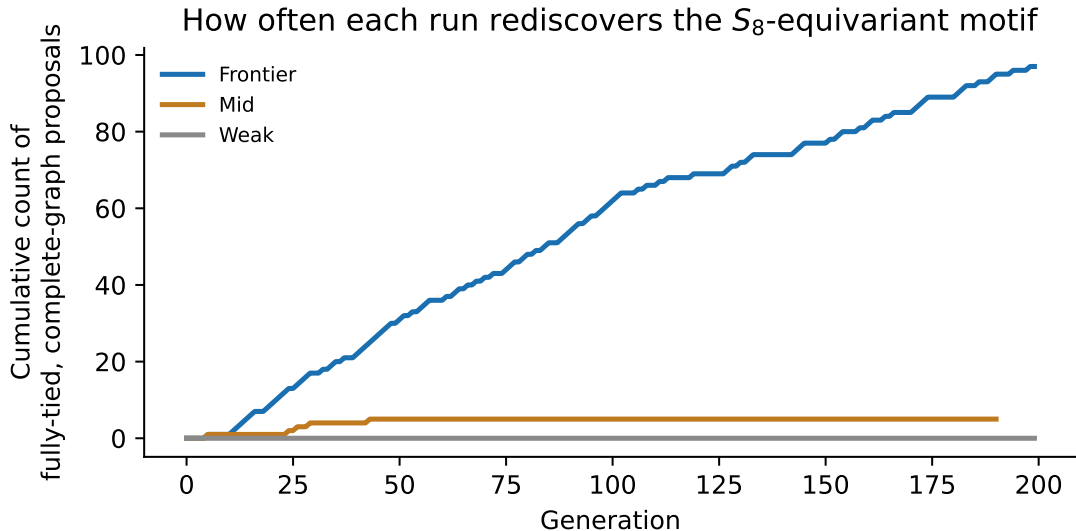


The frontier run’s score jumps from 0.06 (generation 3) and 0.44 (generation 4), both from Mistral-Large-3 tweaking rotation placement and entangler choice, to 0.784 at generation 5 in a single step. Generation 5 was proposed by gpt-5.6-sol, and its own patch description states the reasoning directly:

“Replace wire-specific rotations and the boundary-biased CZ line with a permutation-equivariant block. All qubits share one RY mixing angle, one RZ phase angle, and one final RX readout angle. The complete-graph CZ layer respects the natural symmetry of 28 pair features on 8 qubits...Parameter sharing reduces the total parameter count from 146 to 20.”

This is close to a re-derivation of the hidden structure. Nothing in the prompt says “graph” or “permutation group”; the model reasoned from $28 = \binom{8}{2}$, the one number it was given, to “every pair of qubits is coupled once,” to “the ansatz should treat every wire and every pair the same way,” and produced a complete-graph, fully tied circuit that is exactly S_8 -equivariant on its first attempt at the idea. It even beat our hand baseline’s raw score (0.784 vs. 0.675) immediately, because it also replaced the trailing RZ readout with RX, correctly noting that RZ commutes with the Z-basis measurement and cannot contribute to the observable, a detail our baseline missed. The rest of the run (generations 15, 43, 77, each also by gpt-5.6-sol) is refinement of gate-family assignment within this same equivariant family, not rediscovery of the symmetry itself.

The signal: who finds the motif, and how often



We tagged every proposed program in all three runs with a structural test: does it tie one shared parameter across all 8 single-qubit rotations of a given axis, with no parametrized two-qubit gate breaking the symmetry (matching the shape of generation 5’s design, and of our hand baseline)? The weak run never produces this shape, not once in 198 proposals. The mid run produces it 5 times out of 188, all from DeepSeek-V4-Pro or Mistral-Large-3, each an isolated event that did not get refined further. The frontier run produces it 97 times out of 199, and per-model attribution shows why:

Run	Model	Proposals	Mean score	Equivariant-shaped
Frontier	gpt-5.6-sol	102	0.781	67
Frontier	DeepSeek-V4-Pro	61	0.570	25
Frontier	Mistral-Large-3	31	0.451	5
Mid	gpt-5.4	102	0.042	1
Mid	DeepSeek-V4-Pro	39	0.155	2
Mid	Mistral-Large-3	45	0.332	2
Weak	gpt-5-mini	141	0.467	0

Inside the very same frontier ensemble, with the very same task and the same archive of prior attempts to learn from, gpt-5.6-sol reaches the equivariant shape on two thirds of its own proposals and averages 0.781; Mistral-Large-3, seeing the same archive, only manages it 5 times in 31 tries and averages 0.451. DeepSeek-V4-Pro’s 25 equivariant-shaped proposals in the frontier run, against only 2 in the mid run on the same task, show it can imitate and extend a motif once a stronger peer has planted it in the population, but it never originates the idea in either run. The weak run’s patch notes go the other direction on purpose: several proposals from gpt-5-mini explicitly describe “breaking full symmetry with alternating even/odd parameters” as the design goal, moving away from the answer rather than toward it, and the run never finds anything better than a partially tied ring topology.

Conclusion

Evolutionary search with a frontier model ensemble does not merely match a symmetry-aware hand design, it substantially outperforms one, and general LLM-driven evolution looks like a genuinely useful way to explore this kind of high-dimensional circuit design space. But the more actionable finding is narrower and more useful for future runs: the outcome is not decided by evolutionary search in the abstract, it is decided by whether at least one proposer in the ensemble can look at the problem’s numeric shape and explicitly articulate its symmetry. That signal is absent in the weak run, appears a handful of times by accident in the mid run without ever compounding, and appears in the frontier run as a named, reasoned, one-shot design decision that the rest of the run spends its remaining generations refining. Ensemble composition, specifically including one model capable of this kind of structural inference, matters more than raw generation count.

Data: transfer-sn/results_az_frontier_r1, results_az_mid_r1, results_az_weak_r1 (each a programs.sqlite). Baseline and reproduction script: transfer-sn/baseline_program.py, run_seed_vs_baseline.py. Full seed-vs-baseline writeup in transfer-sn/SEED_VS_BASELINE_REPORT.md.